

CSE5026: Introduction and Frontiers of Cognitive Science

Language (II)

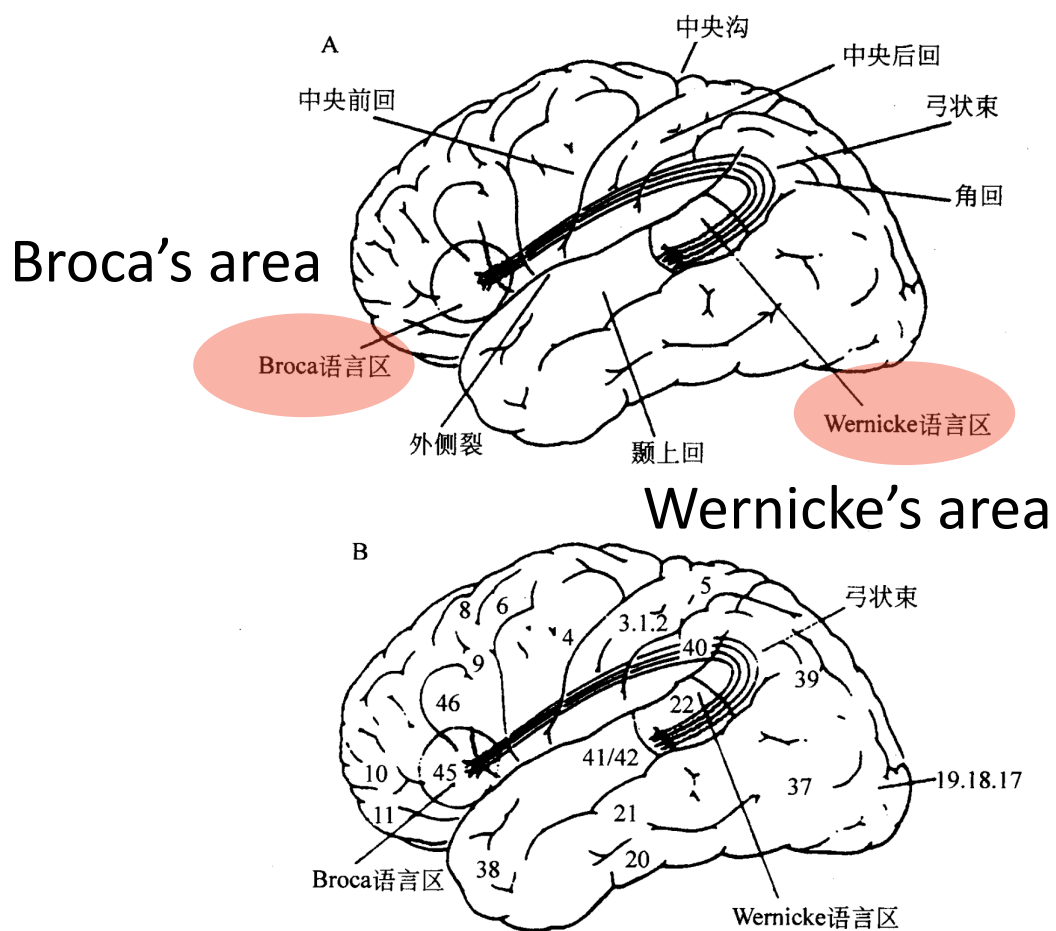
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Overview

- Language Model
- Cognitive Effort in Processing Language
- Dialogue

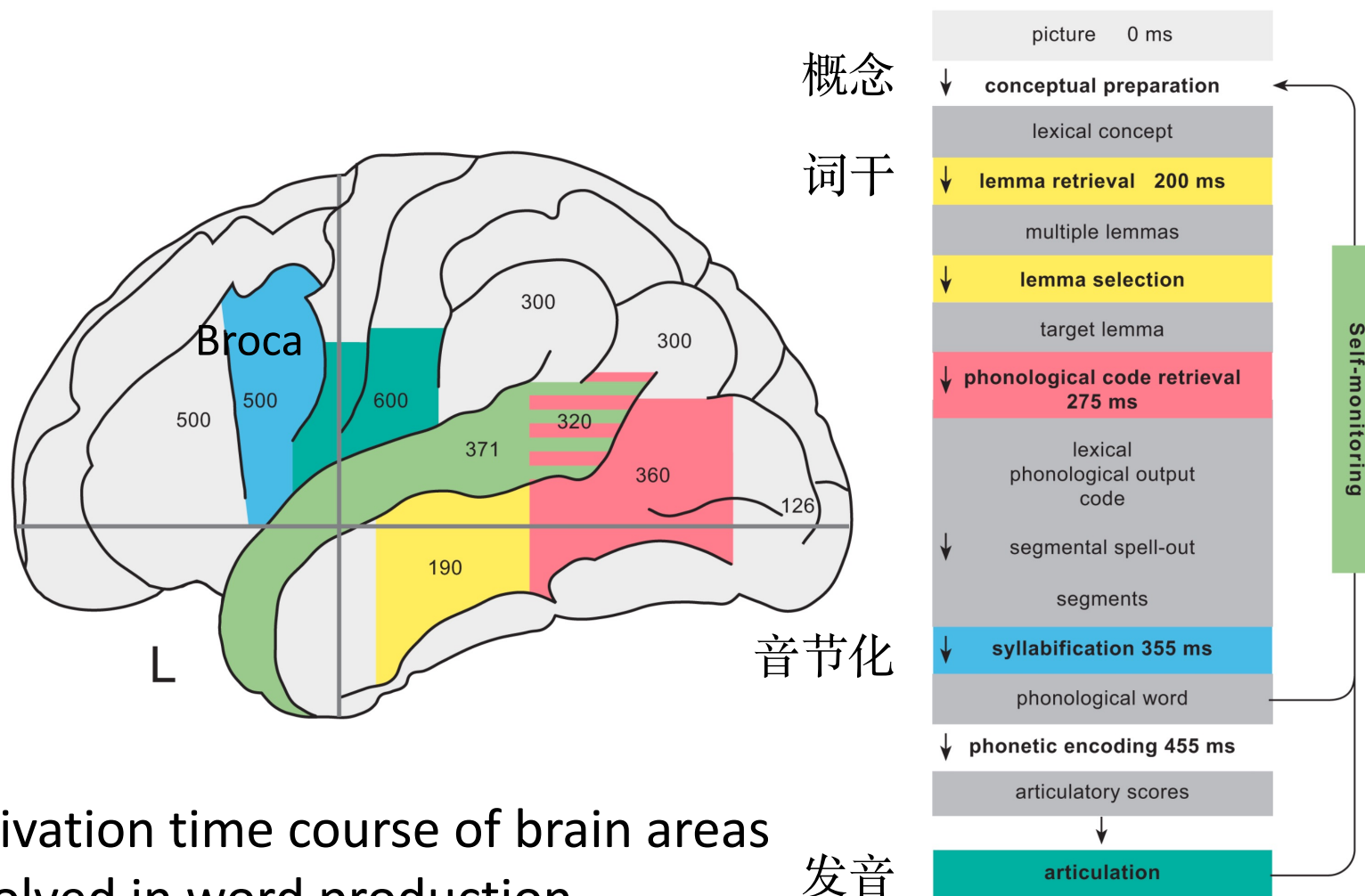
Recap: Brain and Language Comprehension



- 声音信号在听觉系统被转换，信息传递到以角回为中心的**顶颞枕**联合皮质，再传递到 **Wernicke**区，在这里从语音信息中提取出**单词表征 (word representation)**。
- 信息流从 Wernicke 区经过**弓形束 (白质神经束)**到达 Broca 区，这里是**语法特征 (syntactic representation)**记忆之所。
- 接着单词表征激活概念中心相关的**概念 (concept)**。

这样，听觉理解就发生了。

Brain and Language Production

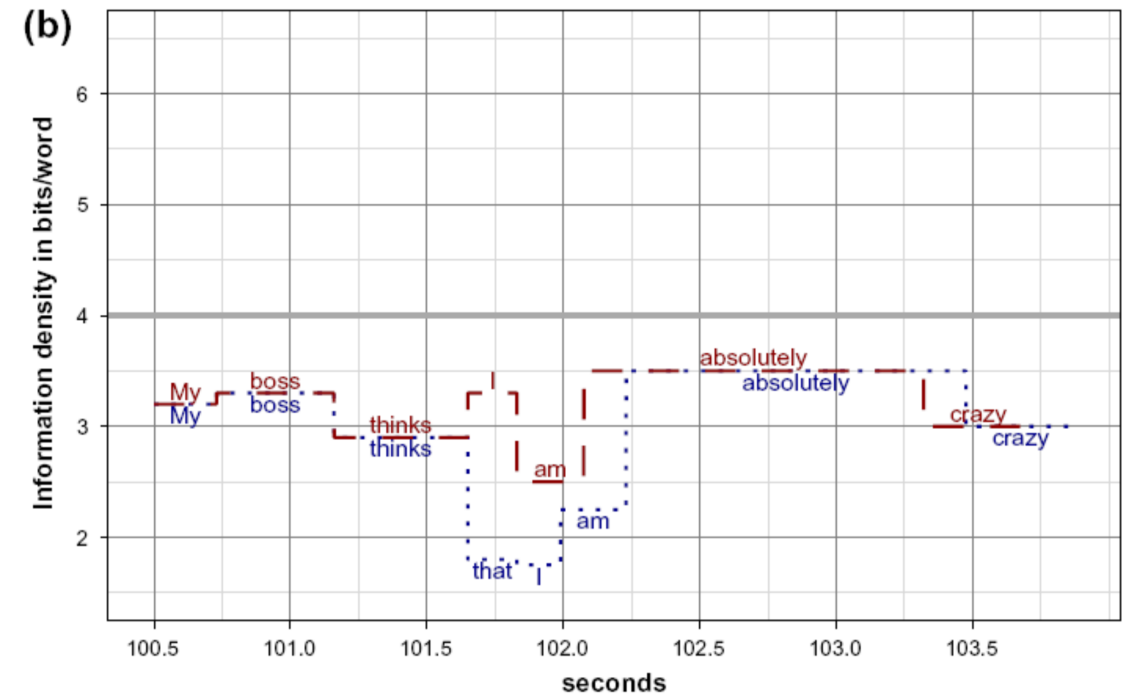
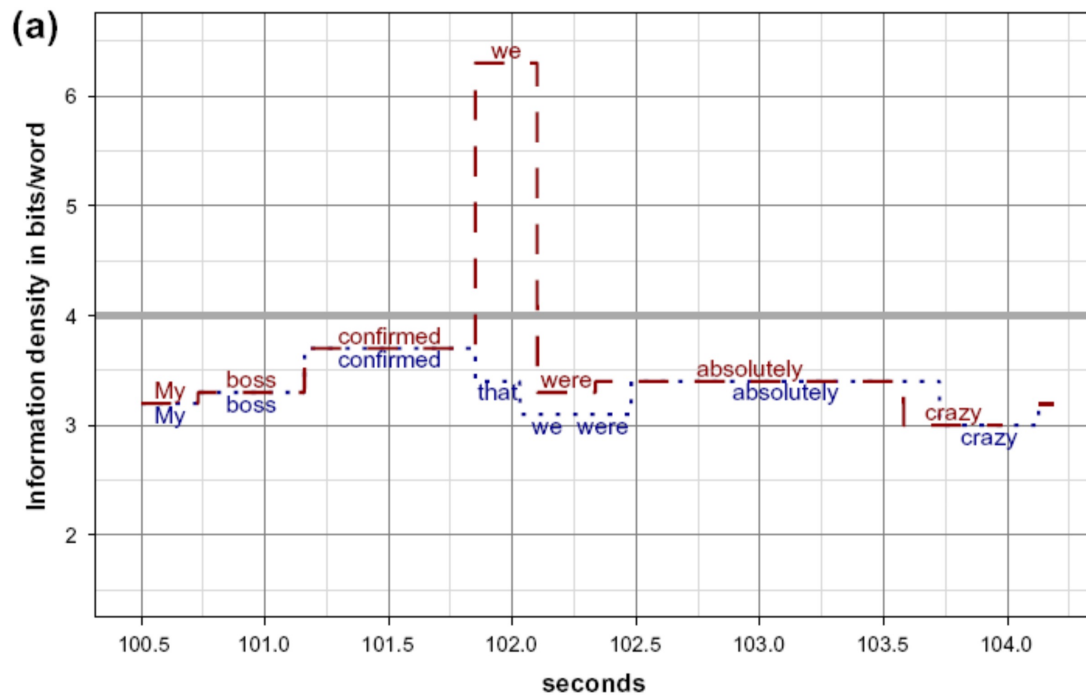


Activation time course of brain areas involved in word production

- **Task:** Naming a picture under fMRI/EEG monitoring
- numbers are time elapse (*ms*) after the onset of stimuli

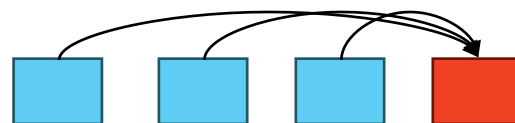
Recap: Information Density

- Uniform Information Density predicts that language production is affected by a (human) preference to distribute information ***uniformly*** across the linguistic signal.

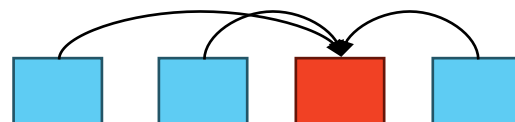


Language Model

- LM: A computational model of predicting words: $w_1, w_2, \dots, w_N$
- Auto regressive LM: $p(w_i | w_1, \dots, w_{i-1}) \Rightarrow p(w_i | w_{<i})$



- Masked LM: $p(w_i | w_1, \dots, \mathbf{w}_{i-1}, \mathbf{w}_{i+1}, \dots, w_N) \Rightarrow p(w_i | w_{\neq i})$



Language Model

- Transformer-based AR LMs are currently the most common type.
- Diffusion-based LMs emerges.

Source: LLaDA <https://arxiv.org/pdf/2502.09992v1>, Dream-7B <https://arxiv.org/pdf/2508.15487>

LM can tell about Information in Language



On the wall I see a X_i

⋮
picture

⋮
pitcher

⋮

X_i can have $|V|$ outcomes, among which

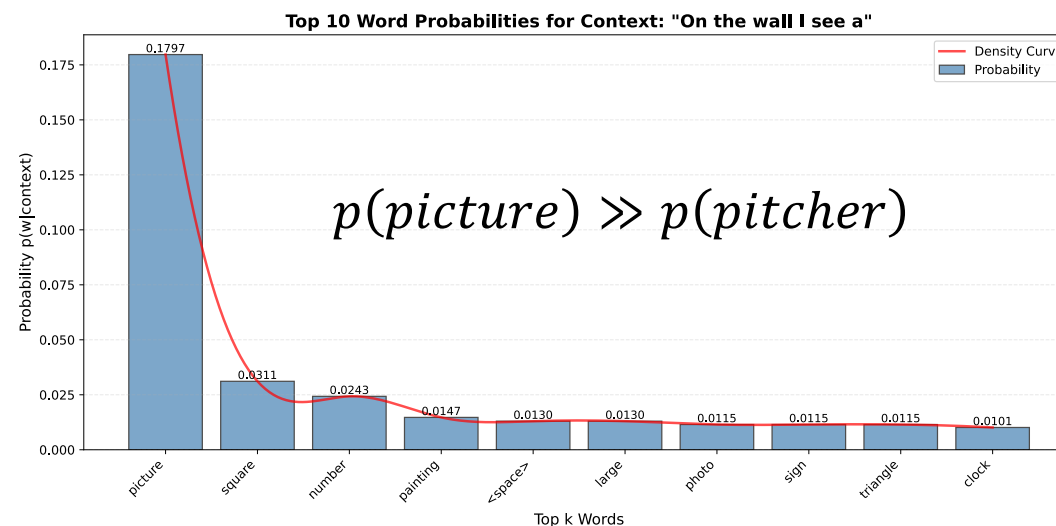
$$p(X_i = \text{picture} | C) = 0.179290$$

$$p(X_i = \text{pitcher} | C) = 6.6 \times 10^{-5}$$

(according to qwen3-9.6B-base)

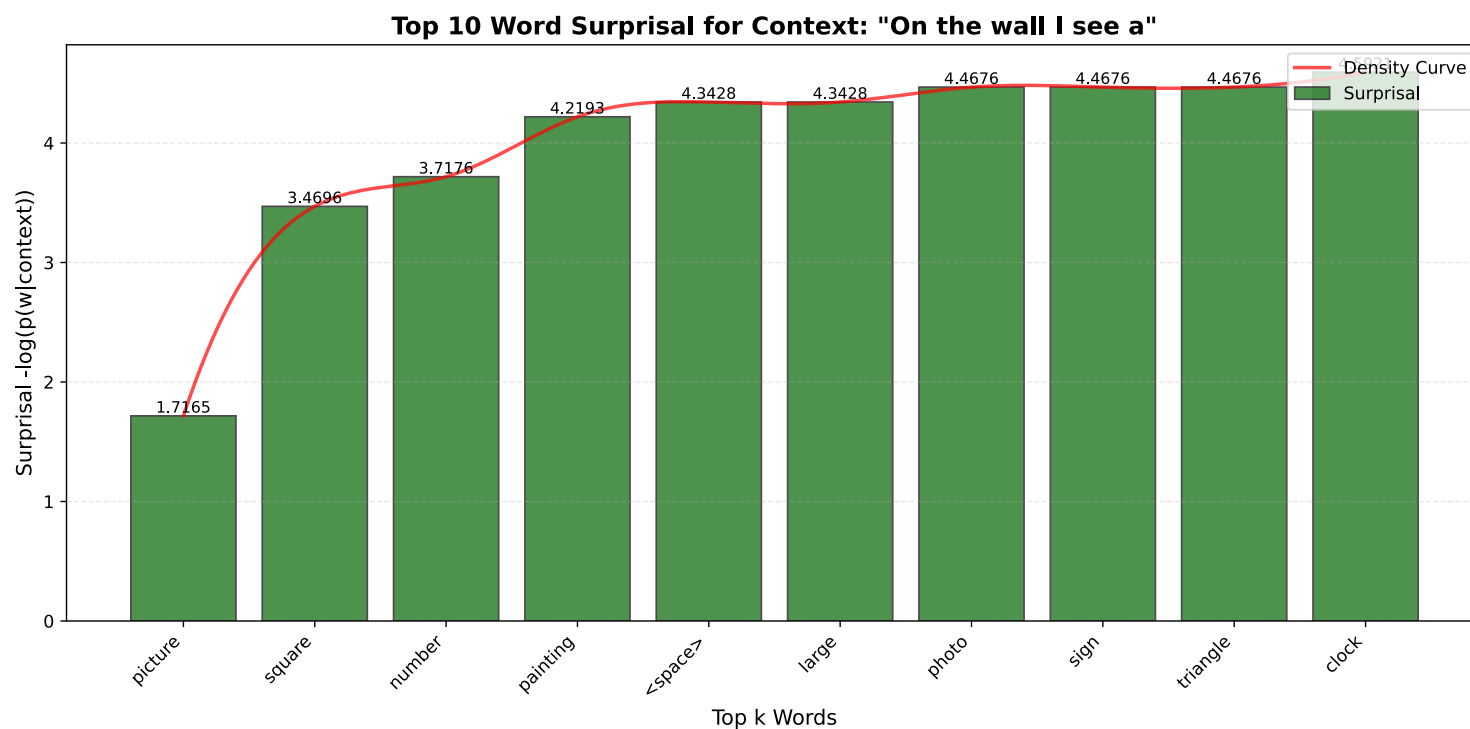
Let's use a discrete random variable X_i to describe the event of generating a word at position i , given its context C .

"picture" is a much more likely outcome of the event



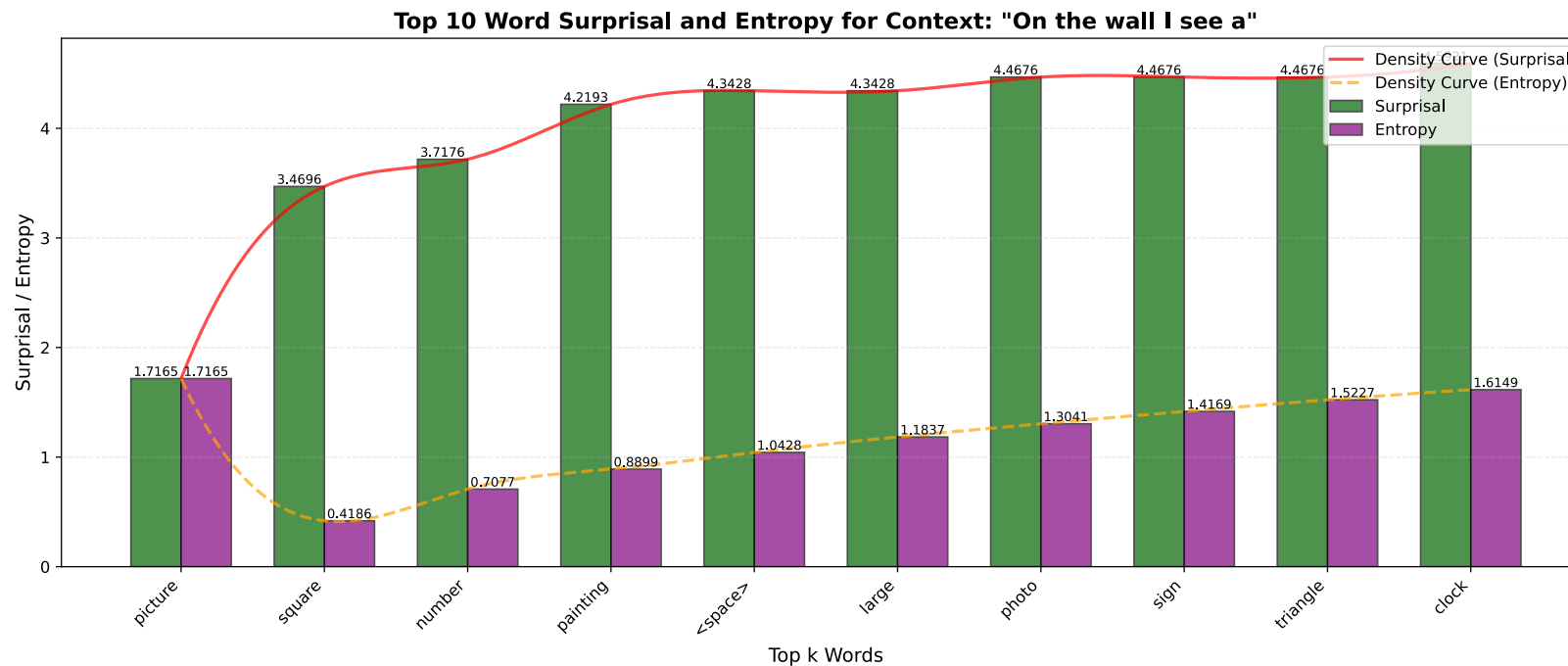
Surprisal of Event Outcome

- Surprisal of an event outcome: $s(x_j) = -\log p(x_j|C)$
 - $s(\text{picture}) = -\log p(\text{picture}|C) = 1.719$, $s(\text{pitcher}) = -\log p(\text{pitcher}|C) = 9.625$



Entropy of Random Variable (Shannon's entropy)

- Entropy of rand. var. X_i , total amount of uncertainty measured by weighted sum of surprisal: $H(X_i) = -p_j \sum_{j=1}^{|V|} \log p(x_j | \mathcal{C})$

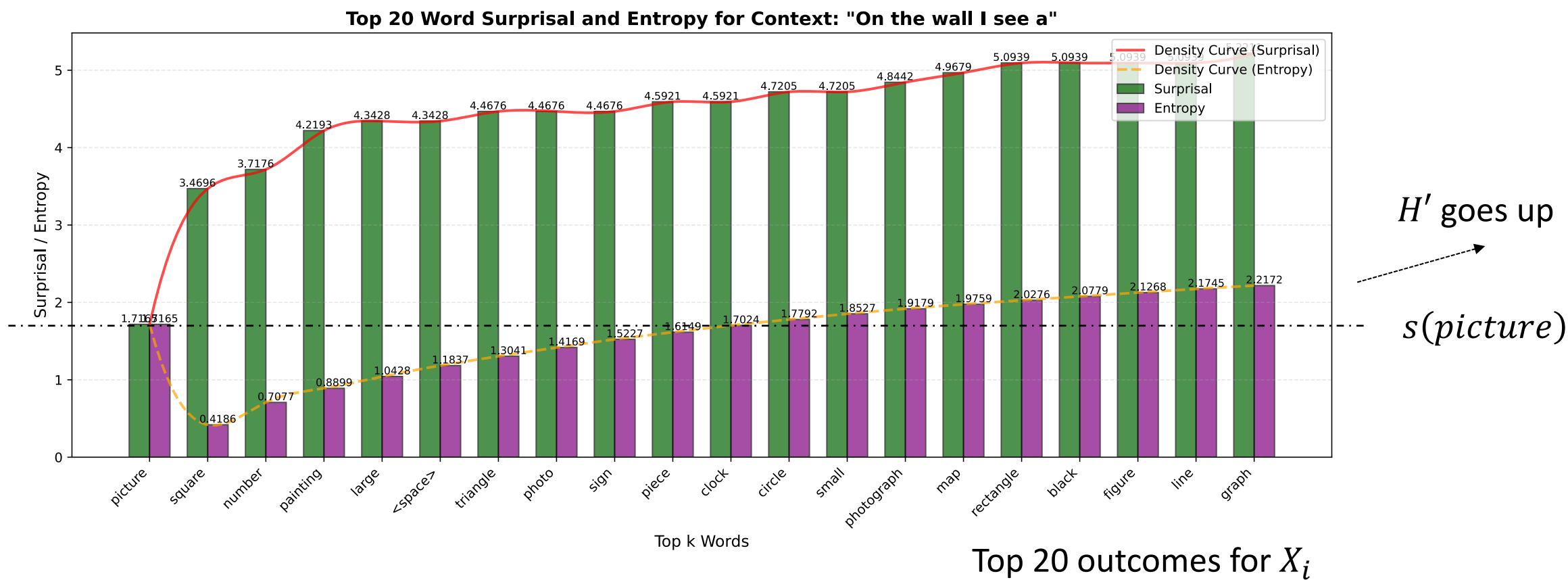


Purple bar (dashed) shows an inaccurate estimation of entropy using the top-k words:

$$H' = -p_j \sum_{j=1}^k \log p_j$$

Entropy of Random Variable

- The empirical values of entropy H are larger than surprisal s



Surprisal can approximate entropy

- Entropy measures the expected surprisal of an unknown event X_i

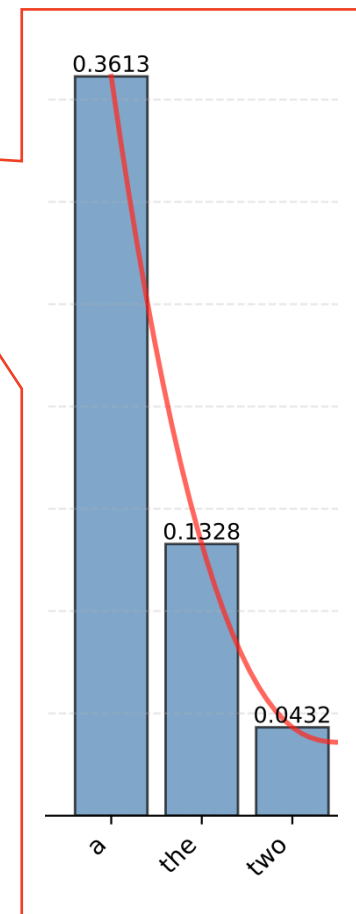
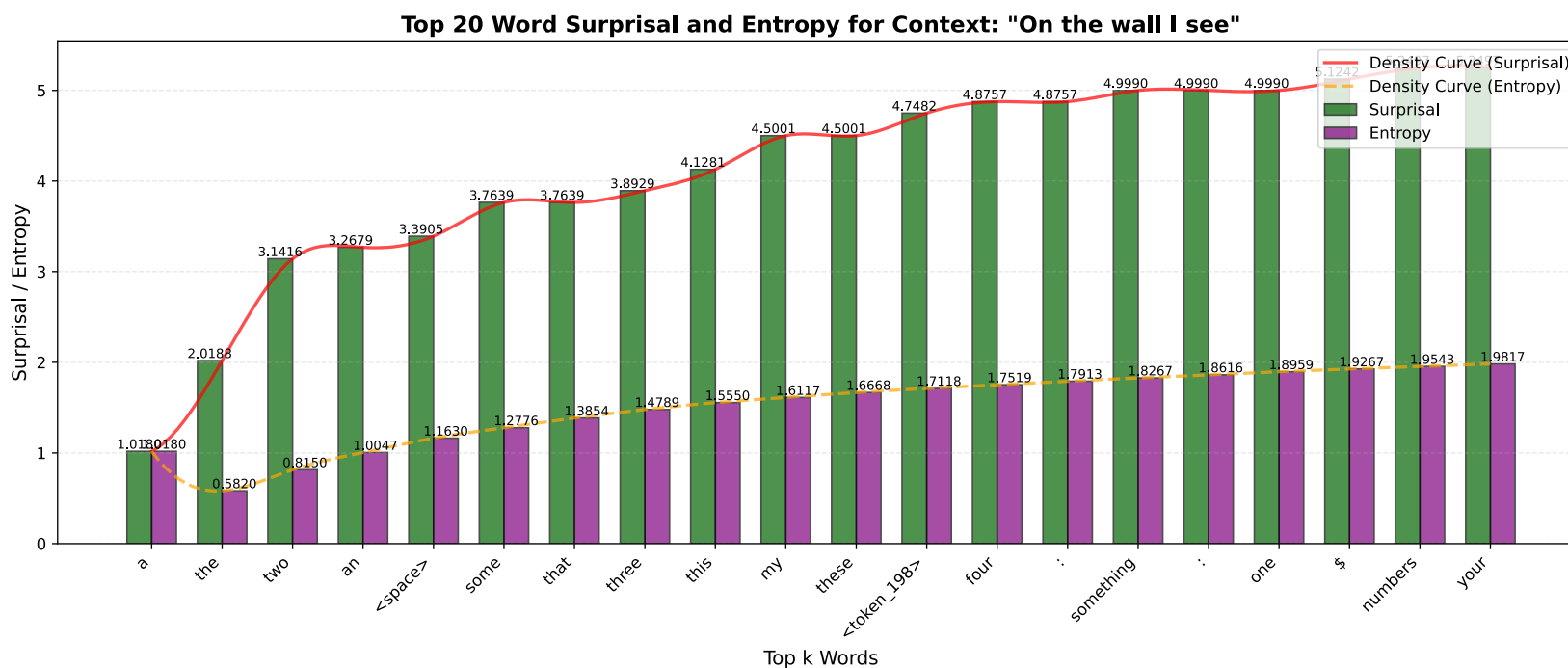

$$H = -\sum_{j=1}^{|V|} p_j \log p_j$$

- Surprisal measures the surprisal of a known outcome:
 $s(x_j) = -\log p(x_j|C)$
- In natural language, the amount of “information” can be defined differently.

Compute entropy for different positions

On the wall I see X_{i-1}

$$\begin{cases} p(X_{i-1} = a|C) = 0.3613 \\ p(X_{i-1} = the|C) = 0.1328 \\ \vdots \end{cases}$$



Measure Information with Entropy Reduction

- Entropy reduction: How much uncertainty is reduced due to the occurrence of a word?







On the wall I see X_{i-1}

$$H'(X_{i-1}) \approx 1.9817$$

using $k = 20$

Negative reduction means the uncertainty increases due to the production of “a”.



$$\Delta H = H'(X_i) - H'(X_{i-1}) = -0.2355$$








On the wall I see a X_i

$$H'(X_i) \approx 2.2172$$



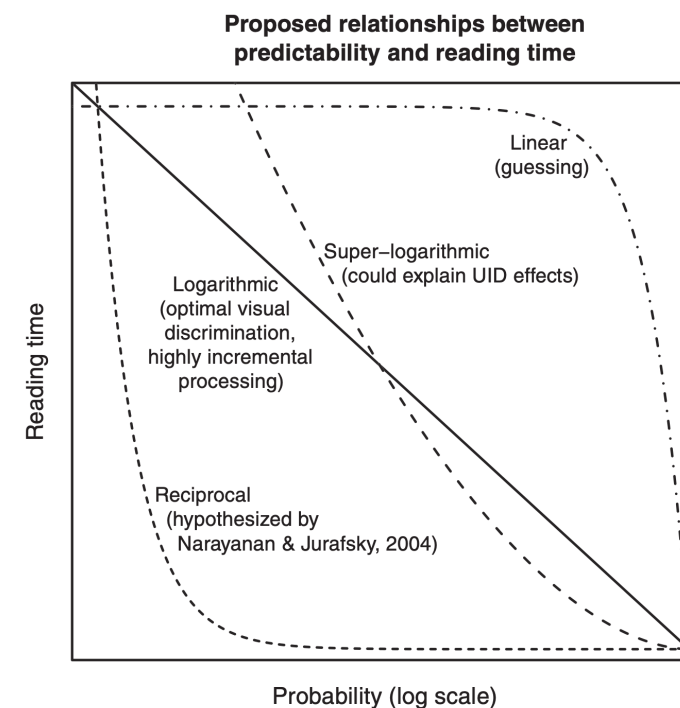
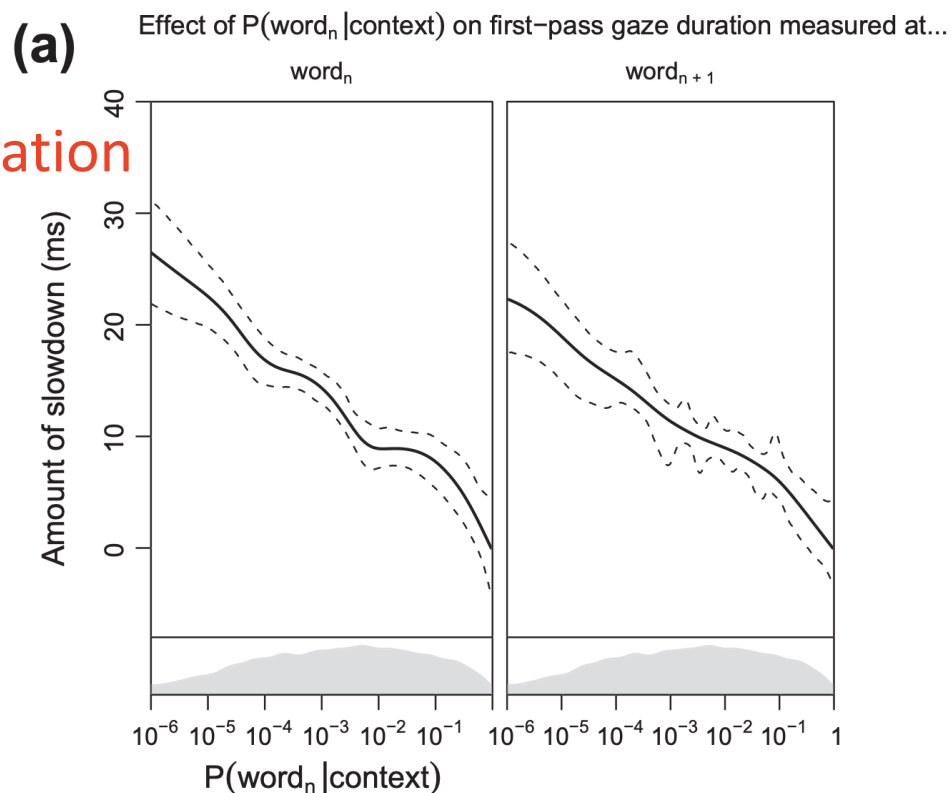
Overview

- Language Model
- **Cognitive Effort in Processing Language**
- Dialogue

Source: Smith and Levy, 2013, <http://dx.doi.org/10.1016/j.bandl.2014.10.006>

Information and Cognitive Effort

- Words of lower probability take more time to read



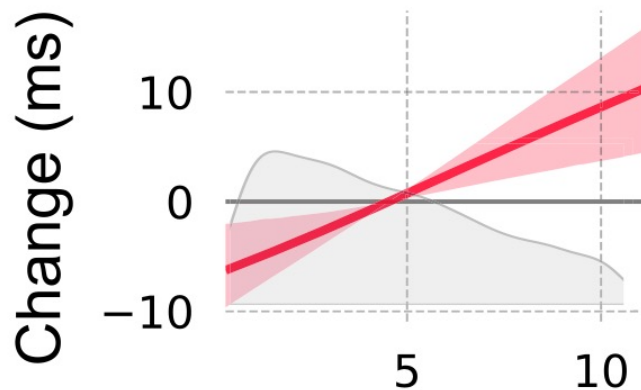
Several different theories to explain the relationship

Source: <https://doi.org/10.1073/pnas.2307876121>

Information and Cognitive Effort

- Word predictability has *logarithmic* effect on reading time.
 - lower probability $p \Rightarrow$ higher surprisal $-\log p$

Relative reading time



$-\log p$ from GPT-2, GPT-J, GPT-3 etc.

Source: <https://acl2025-eyetracking-and-nlp.github.io/>

Are LLMs aligned with Human Reading?

- A research direction: Predictive power of LLM-extracted surprisal (or other metrics) for human reading times (RTs)

Step 1:

Fit two regression models predicting reading times, **with** and **without surprisal as predictor**

Step 2:

Compute Log-Likelihood (LL) of each model

Step 3:

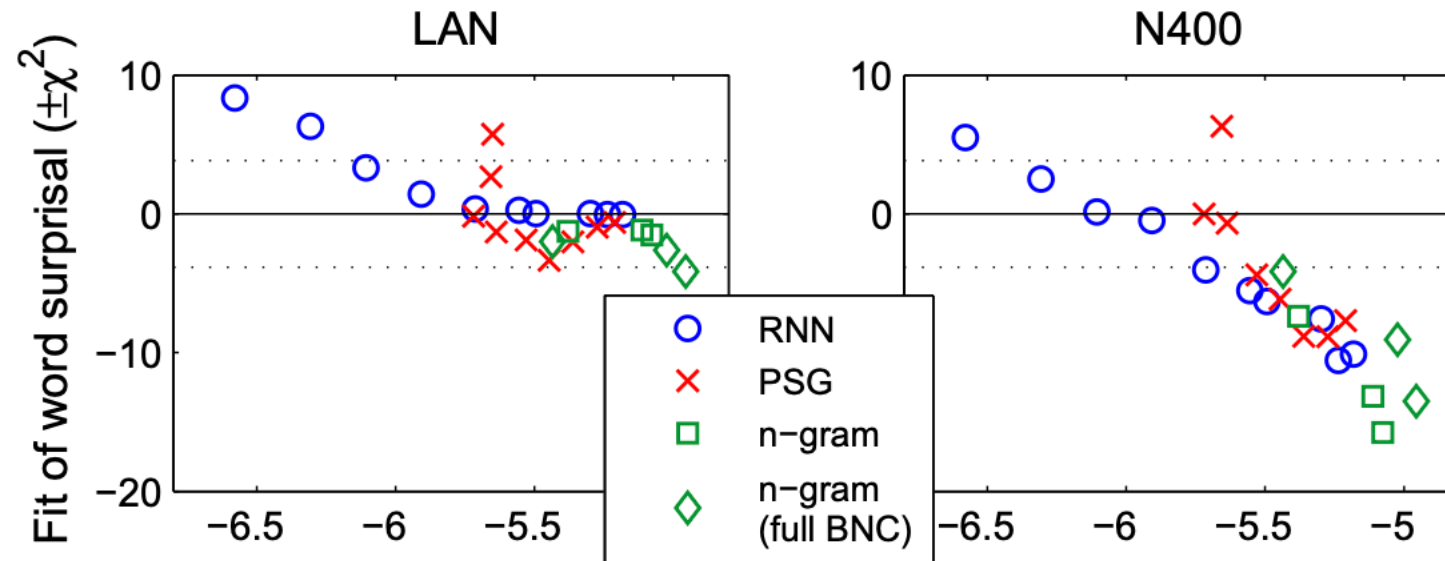
Predictive Power (PP) of surprisal is the ΔLL of the two models

$$\begin{array}{lll}
 \mathcal{M}_{baseline}: RT \sim baseline_variables & \longrightarrow & LL(\mathcal{M}_{baseline}) \\
 \mathcal{M}_{surprisal}: RT \sim baseline_variables & \longrightarrow & LL(\mathcal{M}_{surprisal}) \\
 \oplus surprisal & & \\
 & \longrightarrow & PP(surprisal) \\
 & & := \Delta LL(\mathcal{M}_{surprisal}, \mathcal{M}_{baseline}) \\
 & & := LL(\mathcal{M}_{surprisal}) - LL(\mathcal{M}_{baseline})
 \end{array}$$

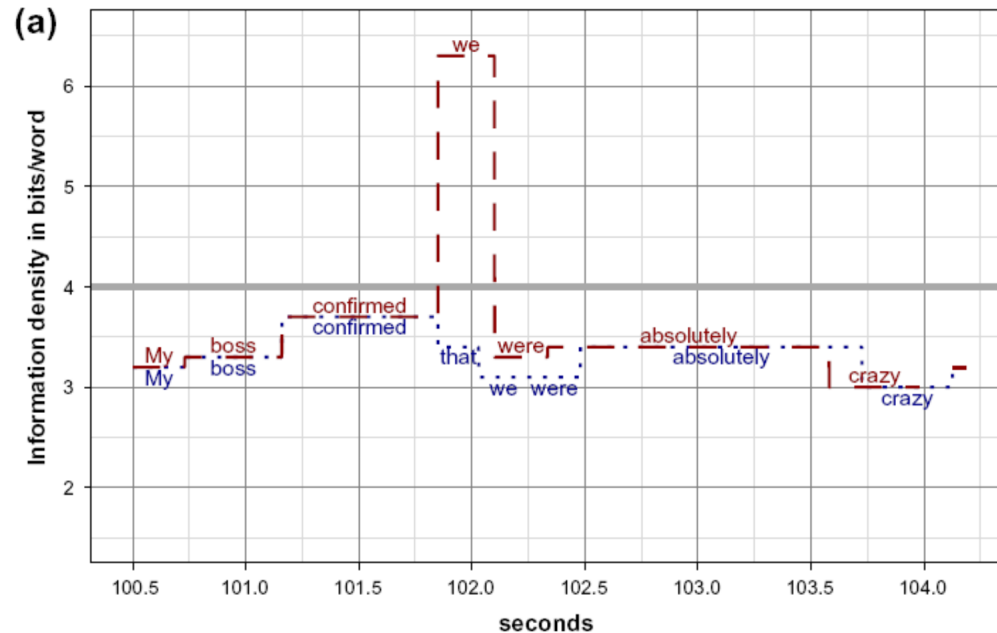
Source: Frank et al., 2015 <http://dx.doi.org/10.1016/j.bandl.2014.10.006>

EEG Evidence

- A comparison between the information measures and ERPs revealed a reliable correlation between N400 amplitude and word surprisal.



Evidence of UID at Corpus Level?



- Human speakers indeed tend to avoid the sudden changes in information in these very specific scenarios.
- How about a more generic trend at corpus level?

Which term is uniformly distributed?

$$H(X|Context) = H(X|C, L) = \boxed{H(X|L)} - I(X; C|L)$$

Approximated by surprisal

- Does $H(X|Context)$ keep relatively constant?
- UID $\Rightarrow$ Entropy rate constancy (ERC) hypothesis.

Approximation due to the limited context

$$H(X|Context) = \underbrace{H(X|C, L)}_{\text{No}} = \underbrace{H(X|L)}_{\text{Yes}} - \underbrace{I(X; C|L)}_{\text{No}}$$

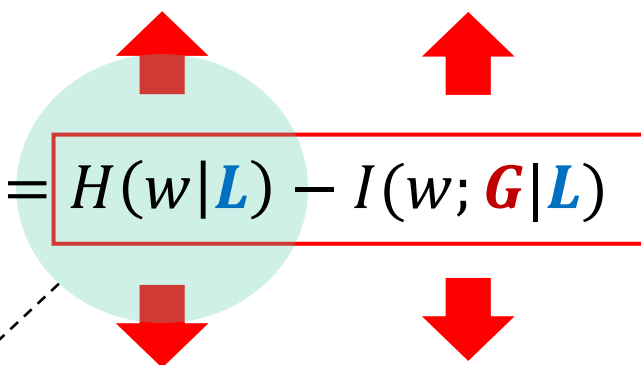
A sentence of N words, $w_1, w_2, \dots, w_N$

$$H(X|L) \approx -\frac{1}{N} \sum_{k=1}^N \log p(w_k | w_{<k}) \quad \text{Average surprisal of all words in sentence}$$



Estimated using LMs

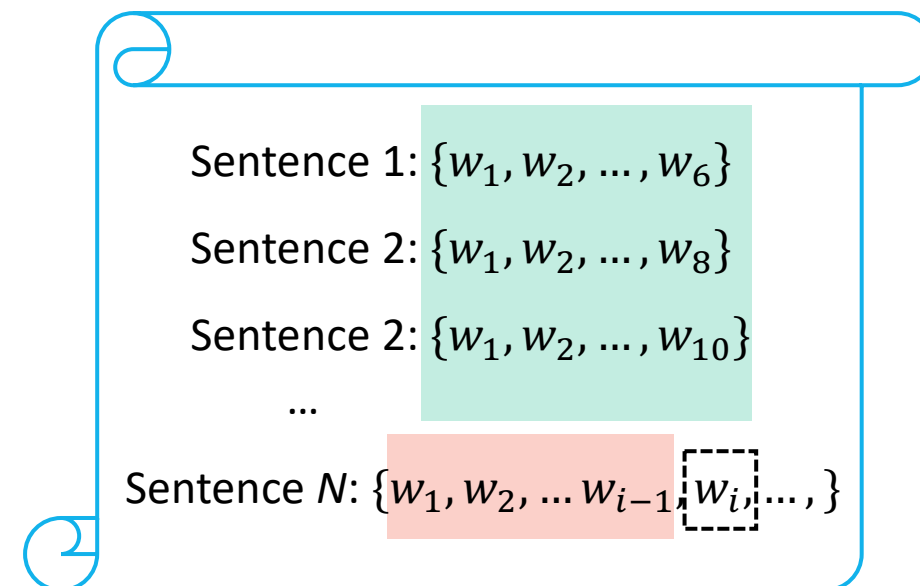
Assuming UID/ERC is true

$$\text{Constant} \leftarrow \boxed{H(w|Context)} = H(w|\mathbf{G}, \mathbf{L}) = \boxed{H(w|\mathbf{L}) - I(w; \mathbf{G}|\mathbf{L})} \rightarrow \text{Constant}$$


Expected observation:

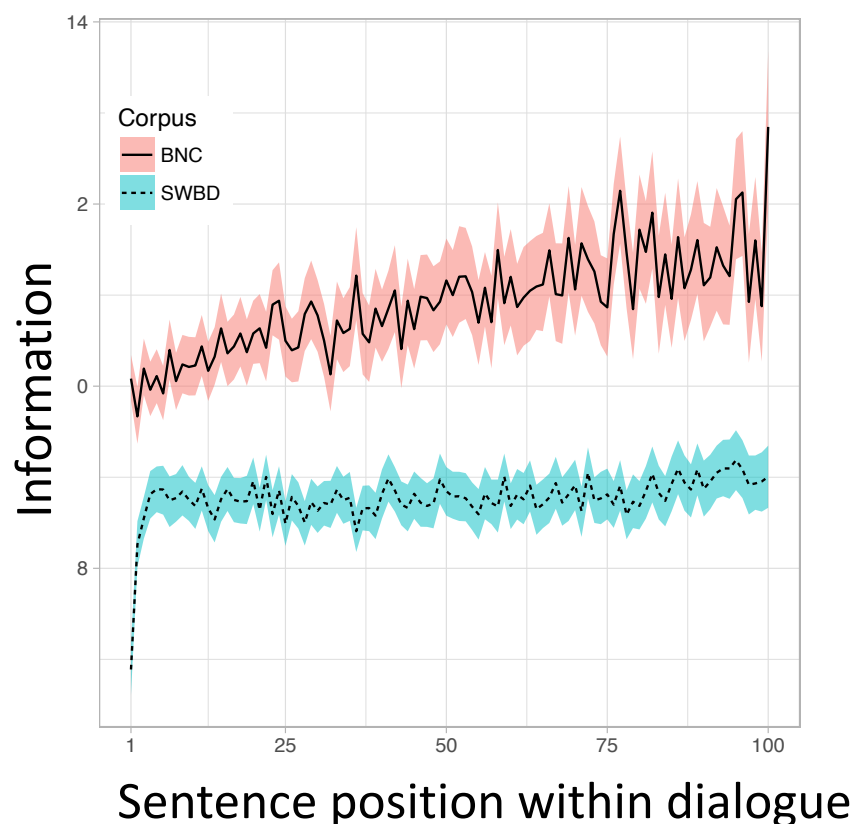
- $I(w_i; G | L)$ increases with i
- $I(w_i; G | L)$ decrease when topic shifts

Surprisal measures the **difficulty** of understanding word w
 Good speakers put hard words later (audience design)



Expected observation 2 confirmed (in dialogue)

- $H(X_i|L_i)$, surprisal, increases with i



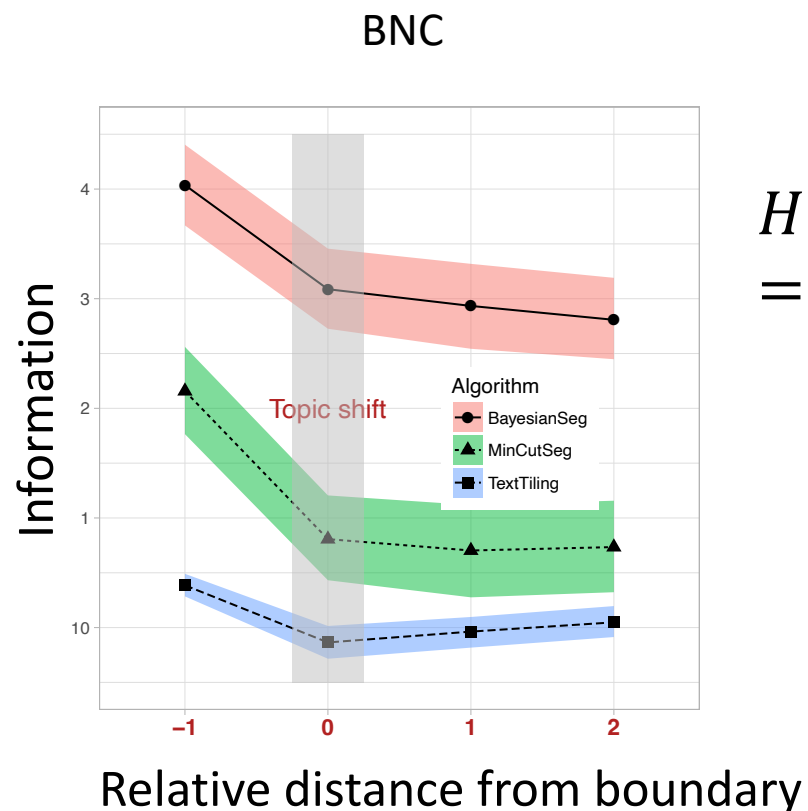
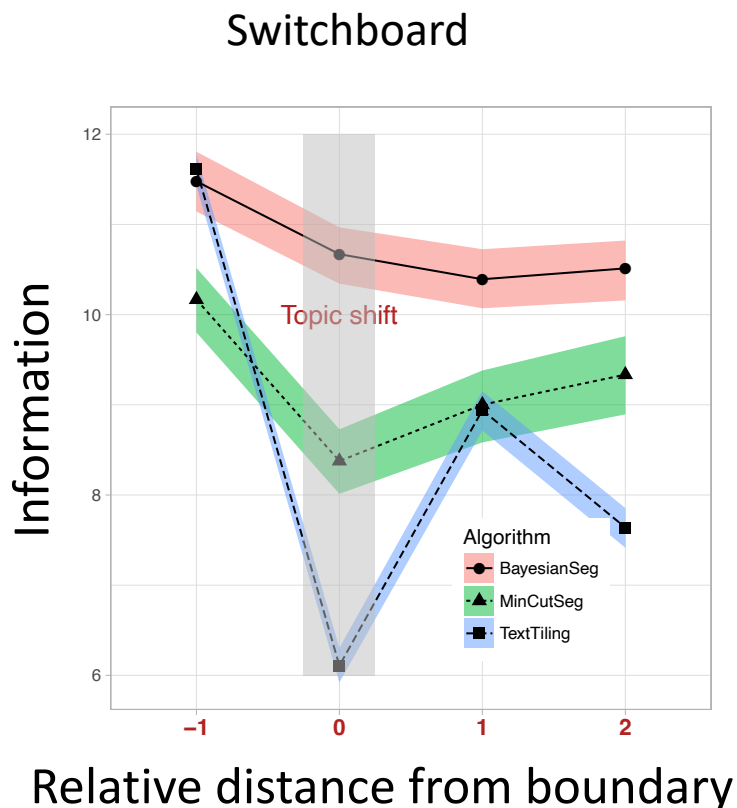
$$H(w|Context) = H(w|\textcolor{red}{G}, \textcolor{blue}{L}) = H(w|\textcolor{blue}{L}) - I(w; \textcolor{red}{G}|\textcolor{blue}{L})$$




Figure from Xu et al., 2016

Expected observation 2 confirmed (in dialogue)

- Information suddenly **decrease** at topic boundaries.



$$H(w|Context) = H(w|\mathbf{G}, \mathbf{L})$$

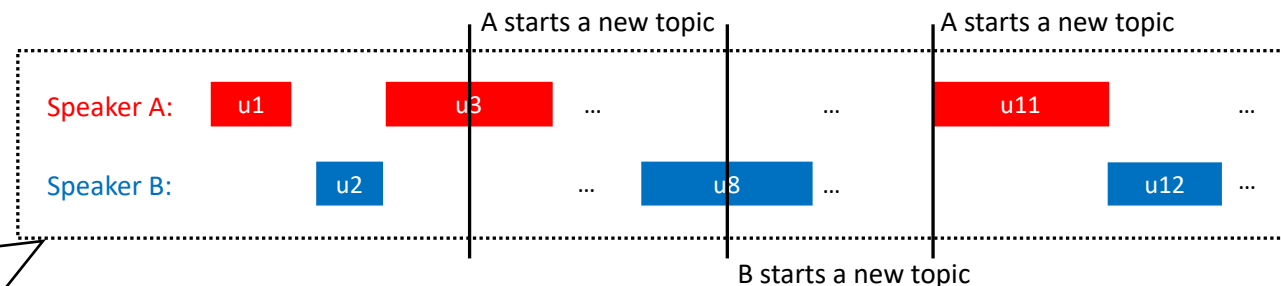
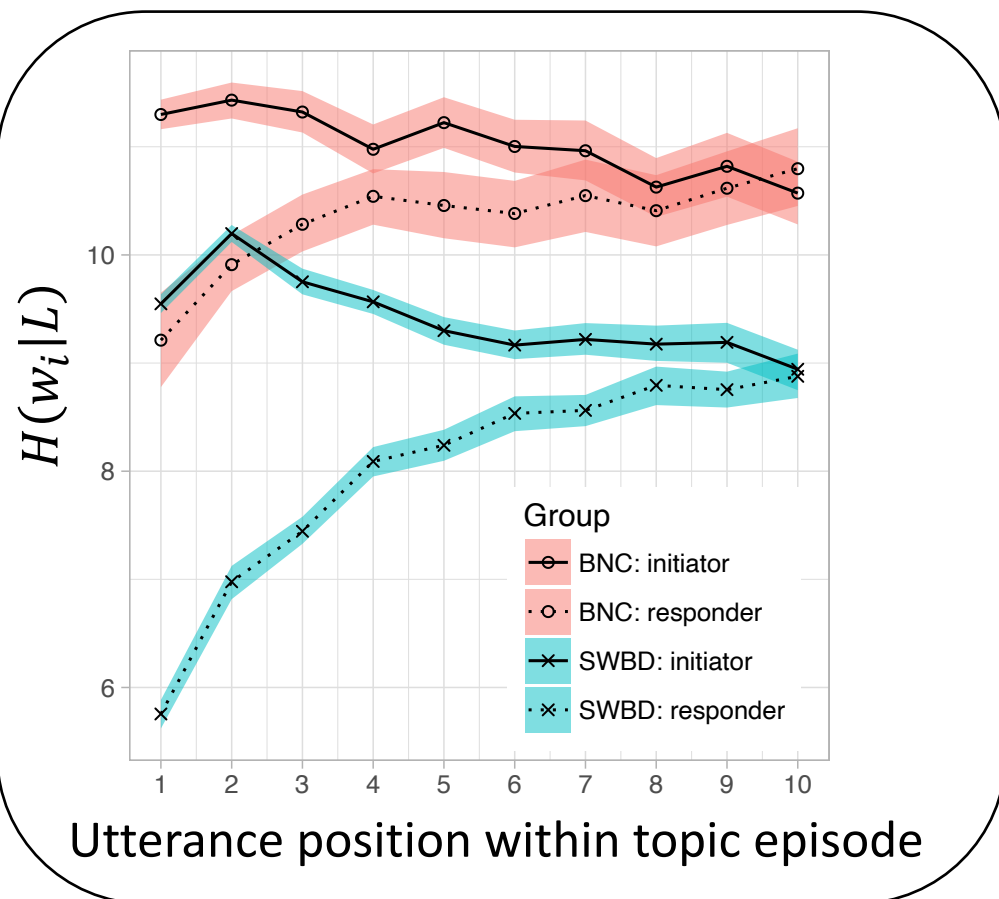
$$= H(w|\mathbf{L}) - I(w; \mathbf{G}|\mathbf{L})$$



Overview

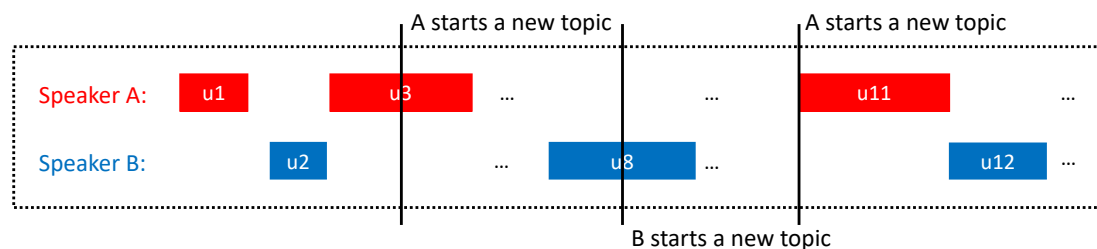
- Language Model
- Cognitive Effort in Processing Language
- **Dialogue**

Surprisal $H(w_i|L)$ converges

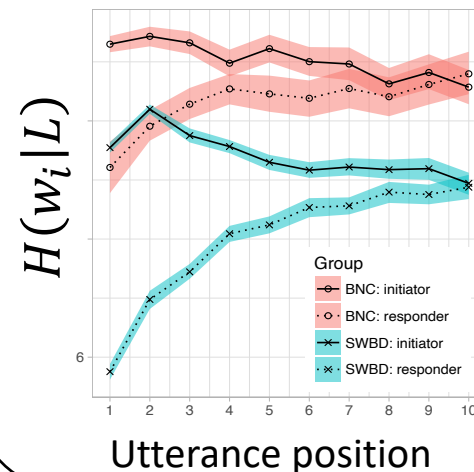


The local entropy (surprisal) of topic **initiators** and **responders** increases form a *convergence* pattern within topic episode

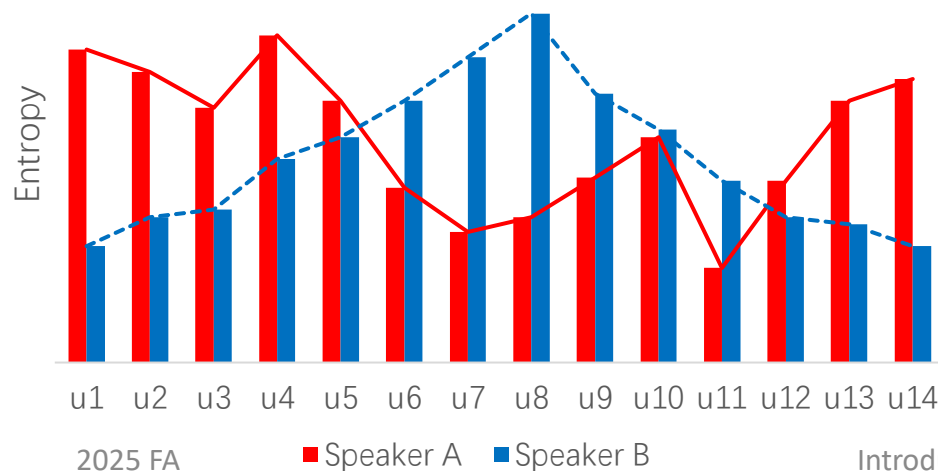
Periodicity of information in dialogue



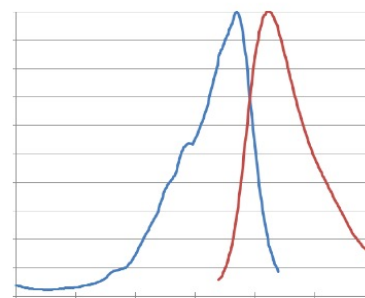
Topic shift occurs **periodically** → Aggregate → Convergence



Time domain



Frequency domain

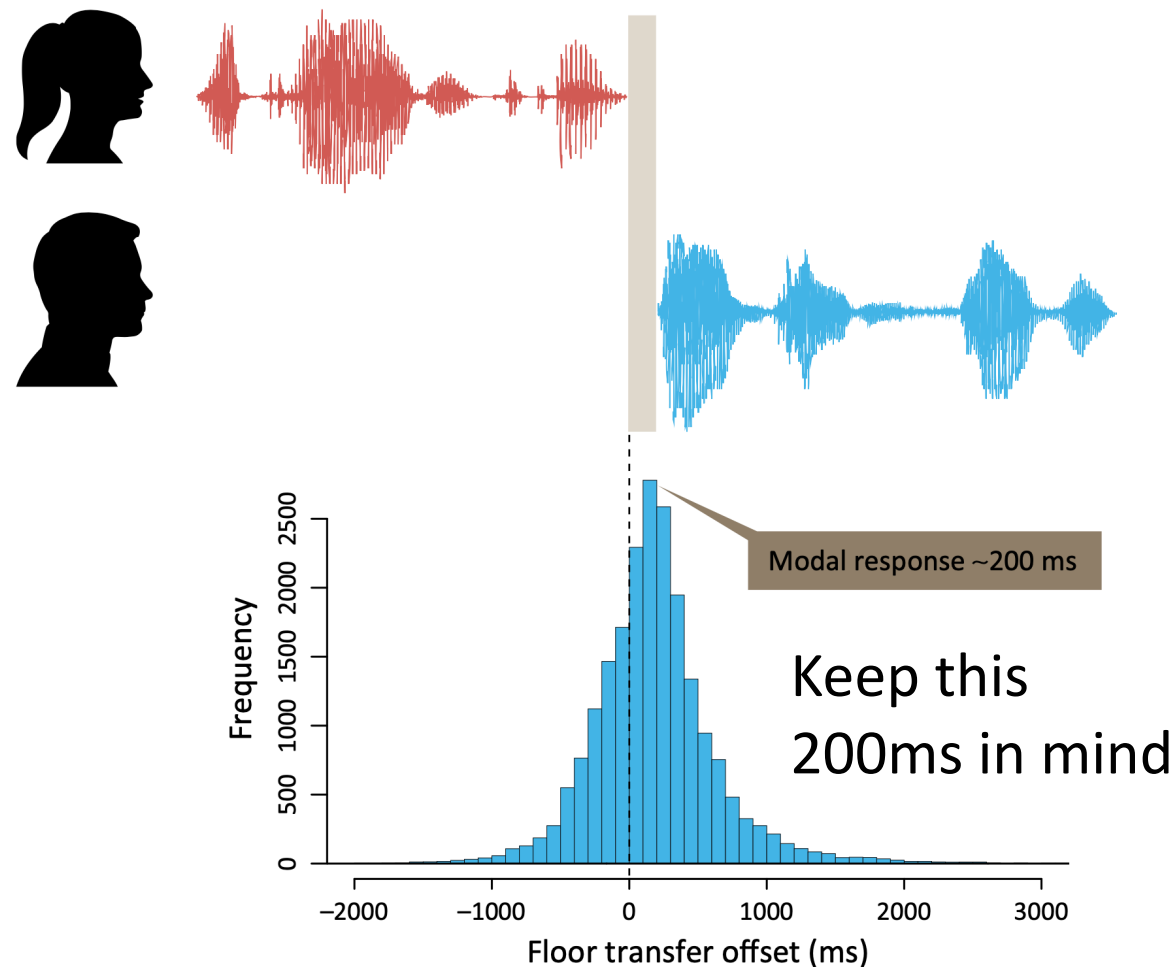


Features from frequency domain may reflect high-level dialogue properties

Turn-Taking 话轮转换

- Language is interactive, involving rapid turn-taking
- Turns are short and responses are remarkably rapid
- Part of a **universal infrastructure** for language

Responses in conversation are fast

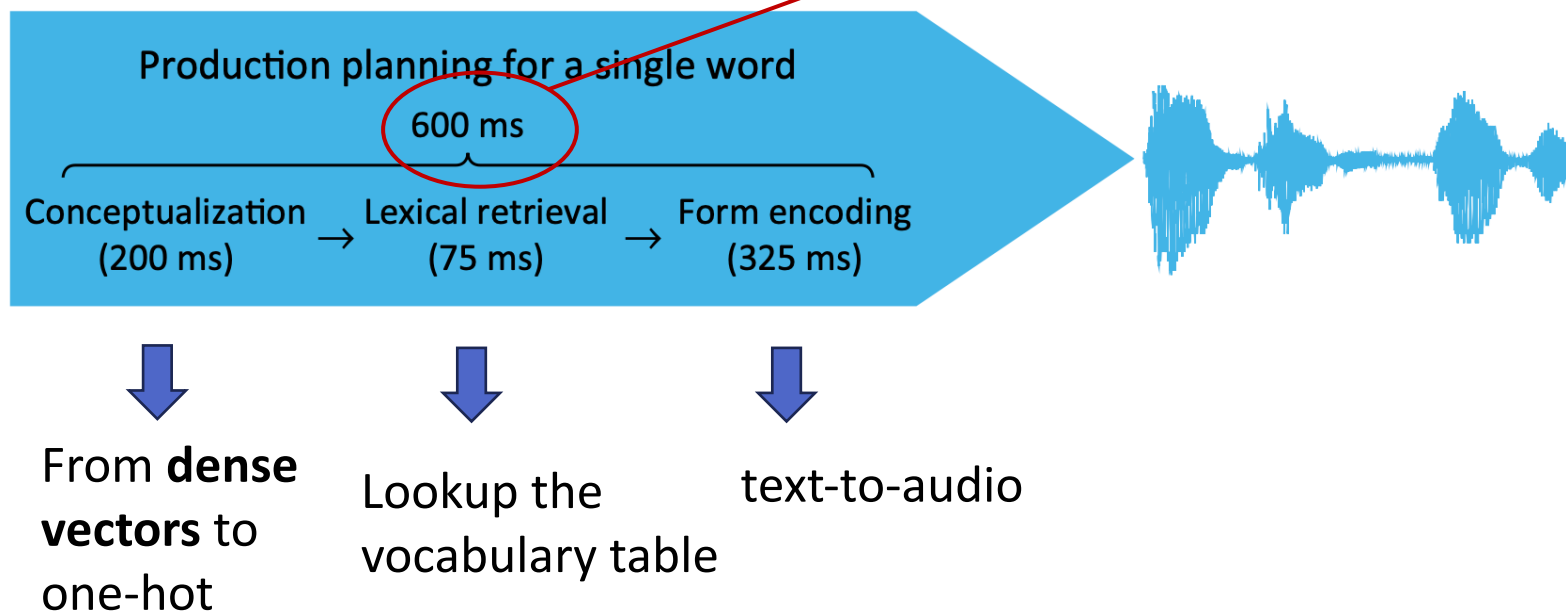


Source: Levinson, 2016

Turn-Taking

Latencies in production are **three fold**:

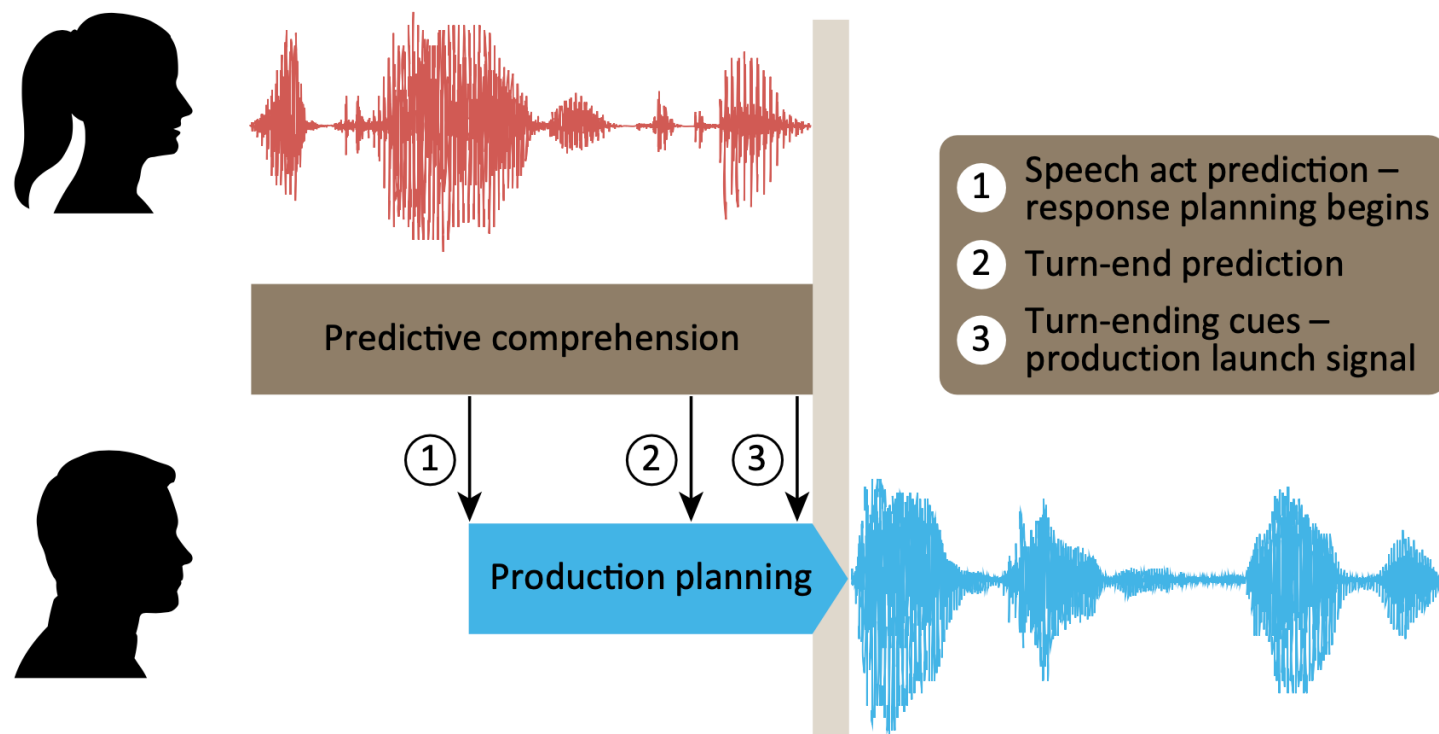
Longer than the **200ms** modal gap!
How do we explain this?



Source: Levinson, 2016

Turn-Taking

Production of response must overlap with comprehension of the incoming turn.



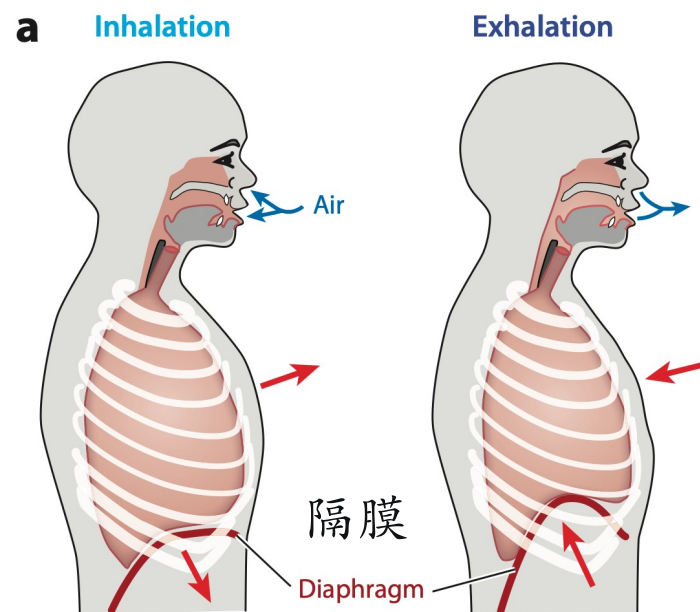
The slow production may be compensated for by predicting the **continuation** or **termination** of the incoming turn, and launching production early

Breathing is related to spoken language

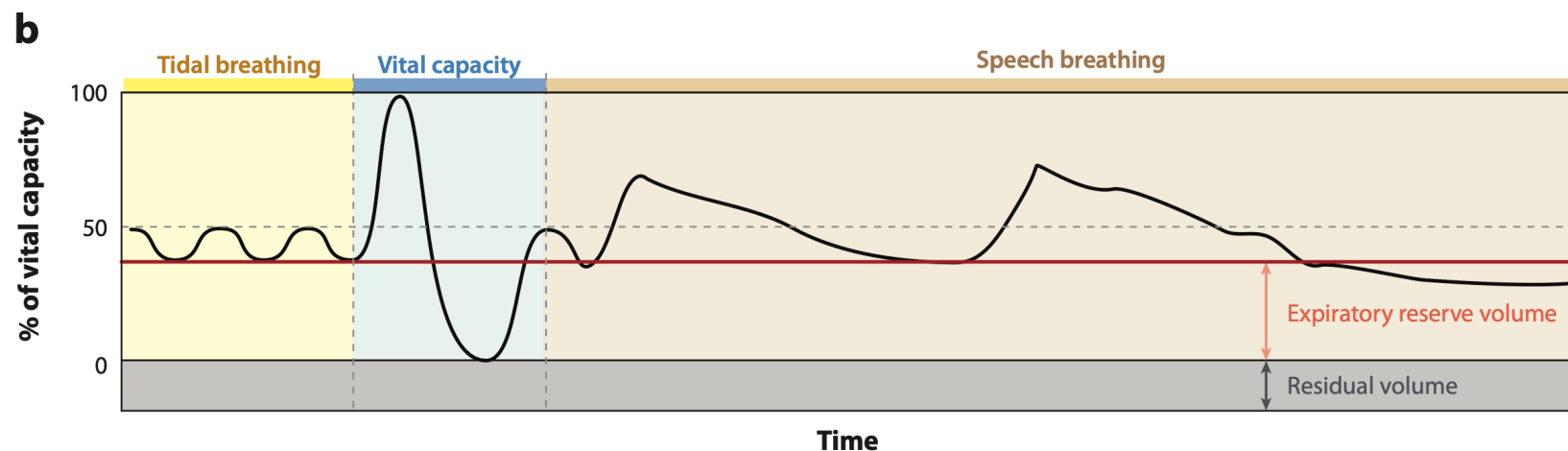
肺活量

Vital capacity: the maximum amount of air that can be breathed out after breathing in as much air as possible.

Speech breathing: involves more variable and deeper inhalations (更深的吸气) than tidal breathing



/ 'daɪəfræm /



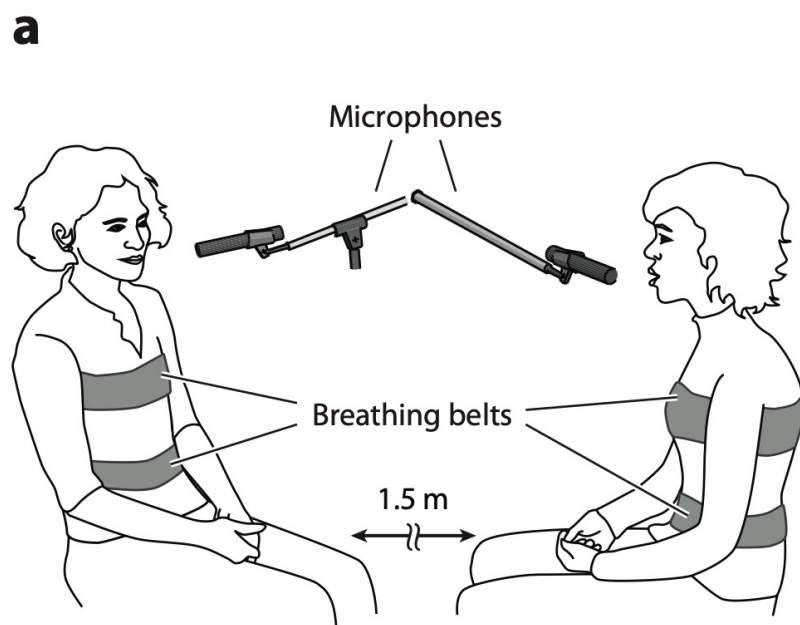
Tidal breathing: the term for inhaling and exhaling while you are relaxing

Speech breathing: short inhalation and a long exhalation

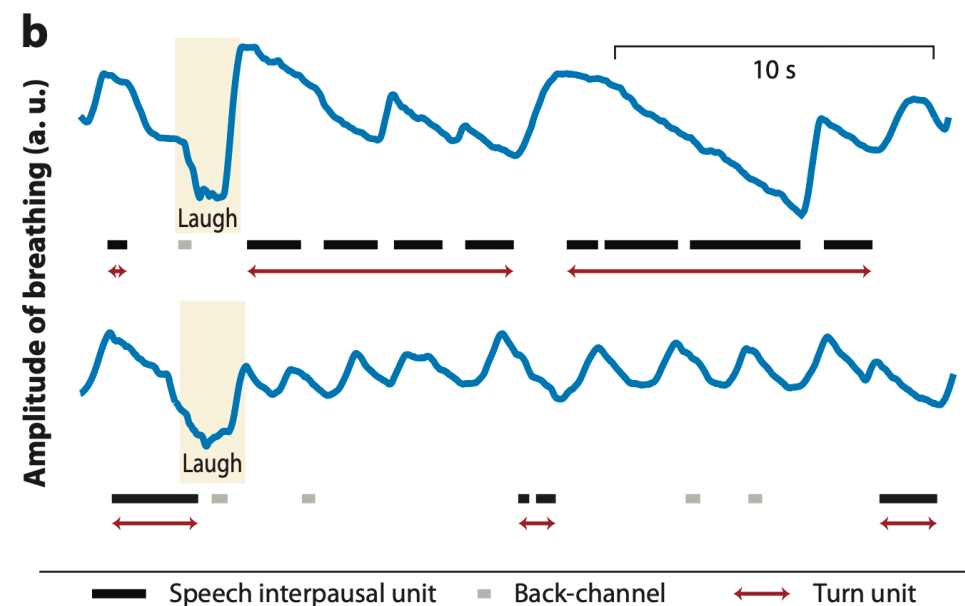
Source: Fuchs and Rochet-Capellan, 2020

Breathing during conversation/dialogue

- Technique: Inductance plethysmography (电感体积描记法)



Breathing signals are recorded using breathing belts



Black bars indicate stretches of speech; red arrow indicates a speaker's complete turn

Breathing adapts to speech

- Inhalation (吸气) at **syntactic boundaries**
 - Rather than occurring randomly in the speech stream, inhalation very often coincides with constituent boundaries
- Anticipating the length of the upcoming sentence
 - Deeper inhalation for longer sentences than for shorter ones

